



Education Algorithm
LEARN • BUILD • LAUNCH

COURSE SYLLABUS

Data Science with Generative AI Roadmap

A structured 3-month program — from Python and statistics to machine learning, deep learning and generative AI / LLM applications.

12

WEEKS

8

STAGES

5

PROJECTS

1

CAPSTONE

Python

Statistics

Pandas/SQL

Machine Learning

Deep Learning

Generative AI

LLMs

- **Beginner-friendly start**
No prior coding or data experience required
- **Project-based learning**
5 hands-on projects, including a GenAI capstone
- **Industry-relevant stack**
The exact tools modern data & AI teams use today

FINAL CAPABILITY

Build and deploy a complete data science pipeline — from data analysis to a working generative AI application powered by LLMs.



The 8-Stage Learning Journey

Each stage builds directly on the last, moving from core data skills to a complete generative AI skill set.

1	Python + Programming	Write programs and work with data structures
2	Statistics + Probability	Apply statistical reasoning to real data
3	Data Analysis + Pandas	Clean, transform and analyze datasets
4	Visualization + SQL	Query data and communicate insights visually
5	Machine Learning	Build and evaluate predictive models
6	Deep Learning + NLP	Build neural networks and language models
7	Generative AI + LLMs	Apply prompt engineering, RAG and LLM APIs
8	Projects + Deployment	Build and deploy portfolio-ready AI applications

FINAL CAPABILITY

By the end of this roadmap, students can build a complete data science and generative AI pipeline — from data analysis and machine learning through to a deployed, presentable LLM-powered application.

Three Months, Three Milestones

MONTH 1

Python & Data Foundations

Python programming, statistics, data analysis basics.

MONTH 2

Machine Learning & Visualization

EDA, SQL, visualization, supervised & unsupervised ML.

MONTH 3

Deep Learning & Generative AI

Neural networks, NLP, LLMs & RAG.

Career-Ready Learning Model

1

LEARN

12 weeks structured path

2

PRACTICE

Concept-first teaching approach

3

BUILD

4

DEPLOY

Deployed capstone project

5

PRESENT



MONTH 1

Python & Data Foundations

Week 1 — Python Programming Foundations

- Python syntax & basics
- Variables, data types, control flow
- Functions & modules
- Practice problems

Week 2 — Data Structures & Libraries

- Lists, dicts, sets, tuples
- NumPy arrays & operations
- Pandas Series & DataFrames
- Reading/writing data files

Week 3 — Statistics & Probability

- Descriptive statistics
- Probability distributions
- Hypothesis testing
- Correlation & regression basics

Week 4 — Data Cleaning & EDA

- Handling missing data
- Outlier detection
- Exploratory data analysis
- Feature engineering basics

MONTH 2

Machine Learning & Visualization

Week 5 — Data Visualization

- Matplotlib & Seaborn
- Chart types & storytelling
- Dashboards with Streamlit
- Visual best practices

Week 6 — SQL for Data Science

- Database fundamentals
- SQL queries & joins
- Aggregations & window functions
- Connecting SQL to Python

Week 7 — Machine Learning Fundamentals

- Supervised learning
- Regression & classification
- Train/test split & metrics
- Scikit-learn workflows

Week 8 — ML Practice & Evaluation

- Unsupervised learning (clustering)
- Cross-validation
- Hyperparameter tuning
- Model evaluation & selection

MONTH 3

Deep Learning & Generative AI

Week 9 — Deep Learning & Neural Nets

- Neural network fundamentals
- TensorFlow/PyTorch basics
- Training & optimization
- CNNs & RNNs overview

Week 10 — NLP & Transformers

- Text preprocessing & embeddings
- Transformer architecture
- Pretrained language models
- Fine-tuning for NLP tasks

Week 11 — Generative AI & LLMs

- LLM fundamentals & APIs
- Prompt engineering
- Retrieval-Augmented Generation
- Vector databases & embeddings

Week 12 — Deployment & Capstone

- Model deployment basics
- Building AI-powered apps
- Git and GitHub
- Final capstone & presentation

Tools & Technologies You'll Master

Python

NumPy

Pandas

SQL

Matplotlib/Seaborn

Scikit-learn

TensorFlow/PyTorch

NLP & Transformers

LLMs & Prompting

RAG & Vector DBs

Git & GitHub



Project Roadmap

Hands-on projects reinforce every stage of the curriculum.

MINI 1**Exploratory Data Analysis Report**

Python, Pandas, EDA, Visualization

MINI 2**Predictive ML Model**

Scikit-learn, Regression/Classification

ANALYTICS**Interactive Data Dashboard**

SQL, Streamlit, Visualization

APPLIED AI**NLP Sentiment Analysis App**

NLP, Transformers, APIs

CAPSTONE**Generative AI Assistant (RAG Chatbot)**

LLMs + RAG + Vector DB + Deployment

What Students Should Be Capable Of

- ✓ Write Python programs and work with core data structures
- ✓ Apply statistics and probability to real-world data
- ✓ Clean, analyze and visualize data with Pandas, SQL & Matplotlib
- ✓ Build and evaluate ML models with Scikit-learn
- ✓ Build deep learning and NLP models with modern frameworks
- ✓ Apply prompt engineering and RAG to build GenAI apps
- ✓ Work with LLM APIs and vector databases
- ✓ Use Git/GitHub; build and present a final AI project

Note: Completing the syllabus alone does not guarantee job readiness. Consistent practice, projects, portfolio work and interview preparation should accompany this roadmap.

Who Should Enroll

1**Absolute Beginners**

No prior coding or data experience needed — the roadmap starts from Python fundamentals.

2**Career Switchers**

A structured, job-oriented path into data science and generative AI within three months.

3**Analysts & Developers**

Deepen existing skills with machine learning, deep learning and modern GenAI tools.

Ready to Start Your Data Science Journey?

Email support@educationalalgorithm.com

Website educationalalgorithm.com